

Topic 3:

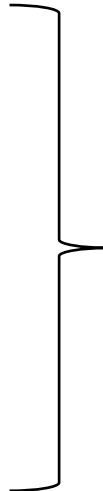
Pressure and Fluid Statics

Part 1: Fluid and Hydrostatic Pressure

INSTRUMENTS & MEASUREMENTS

Topic to cover

- Physical characteristics of fluid
- Mass, weight, density, specific gravity
- Pressure
- Hydrostatic pressure
- Pascal's Principle
- Gauge pressure, absolute pressure
- Pressure measurement
- Archimedes' Principle



Topic 3: Part 1

Physical Characteristics of Fluid

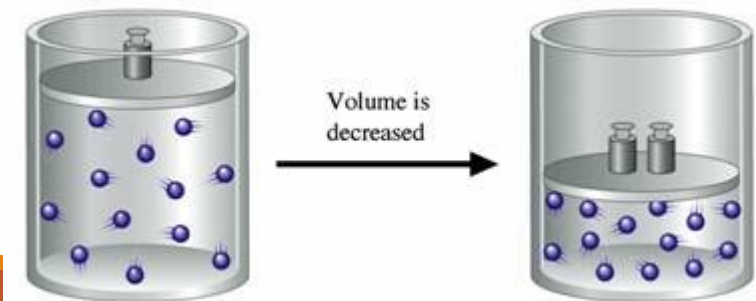
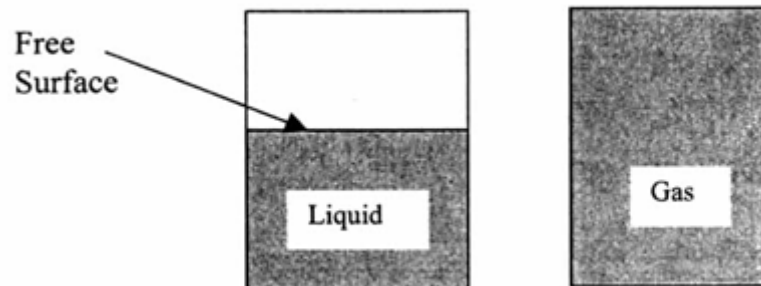
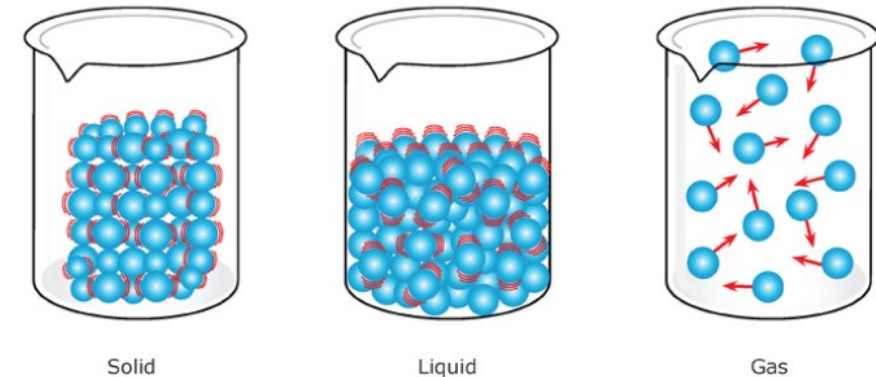
Physical characteristics of fluids

- Both liquid and gas are considered fluids
- Fluid **lacks ability** to **offer permanent resistance** to a **deforming force** (particles can move/slide past one another)
 - **Flow** and **deform continuously** under the **action of force**, causing its shape to change
 - Therefore it can take the **form of any solid body** it **comes into contact with**.



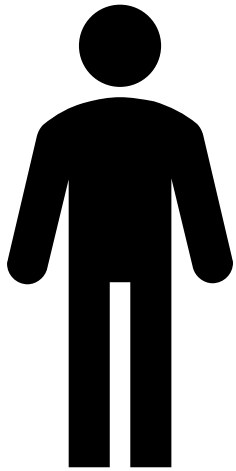
Liquid vs Gas (Fluids)

- Liquid:
 - Almost incompressible (closely held and little free space between particles)
 - Free surface may be formed
- Gas:
 - Readily compressible
 - Does not form free surface



Mass, weight, density

Mass, weight & gravitational pull



My mass is 60kg.

My weight is 588.6N on Earth and 223.2N on planet Mars

Weight (W) = Mass (m) x Gravitation Acceleration (g)

$W = mg$

On Earth, gravitation acceleration is 9.81 m/s²

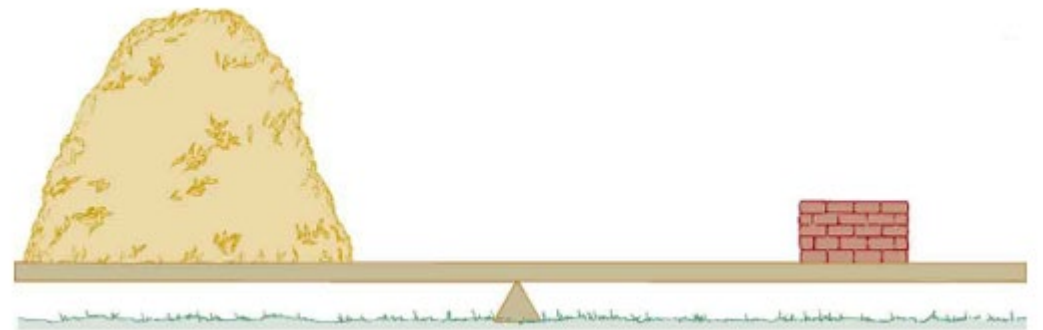
But on Mars, gravitation acceleration is 3.72 m/s²

Note: Weight is a force, so it has a unit of Newton (N)

Density

Which is heavier? A kg of feather or a kg of stone?

Which would occupy a greater volume? A kg of feather or a kg of stone?



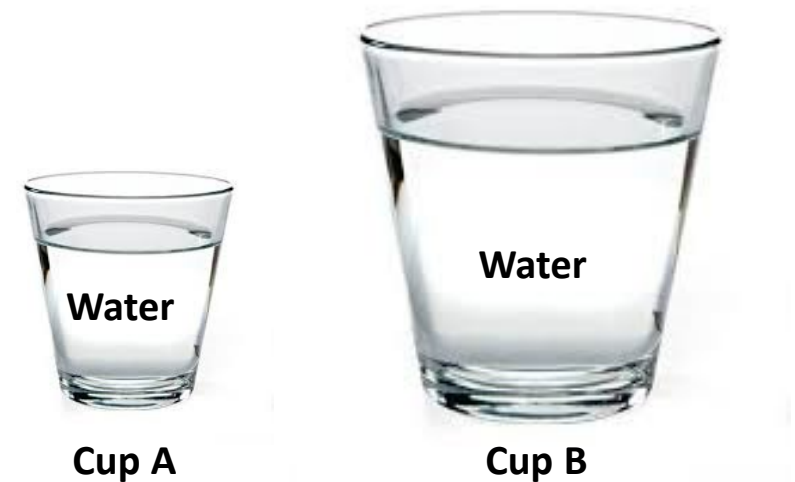
Density

- Mass of a substance per unit volume (ρ)

$$\text{Density} = \text{Mass} / \text{Volume}$$

- Example:
 - Density of water at 4°C = 1000 kg/m³
 - Density of mercury at 20°C = 13546 kg/m³
- Changes with temperature and pressure

Would the density of water be the same for Cup A and B if the temperature and pressure are the same?



Specific Gravity

★ Specific Gravity (S.G.) =
$$\frac{\text{Density of a substance}}{\text{Density of a reference substance}} = \frac{\rho_s}{\rho_{ref}}$$

Where ρ_s = density of substance and ρ_{ref} = density of reference substance

- The most common reference used for solids and liquids is water at 4°C which has the following density:

★ $\rho_{H_2O} (4^\circ\text{C}) = \underline{1000 \text{ kg/m}^3} \text{ or } \underline{1 \text{ g/cm}^3} \text{ or } \underline{62.43 \text{ lb}_m/\text{ft}^3}$

Specific Gravity – using water as reference

$$\text{Specific Gravity (S.G.)} = \frac{\rho_s}{\rho_w}$$

Where ρ_s = density of substance and ρ_w = density of water (the reference substance)

- Shows how many times a substance is denser than water
 - Example: if a liquid has a S.G. = 2, it means it is 2x denser than water
- Has no units of measurement

Example

- $\rho_w = 1000 \text{ kg/m}^3$
- S.G. for mercury (Hg) = 13.54
- *What is the density of Hg (ρ_{Hg})?*

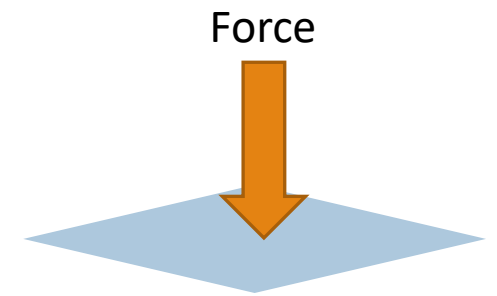
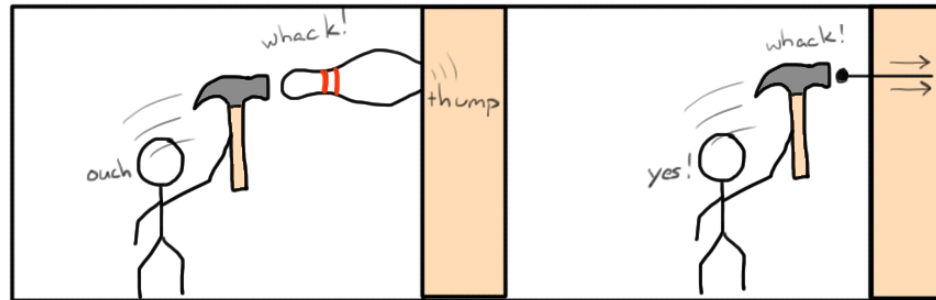
Pressure

Pressure

- Pressure is defined as normal force per unit area.
- Examples of pressure units are N/m^2 and lb_f/in^2 (or psi).
- The SI unit for pressure is pascal (Pa), which is defined as N/m^2 .

Can you tell the difference between force and pressure?

Same force different pressure



A normal force acting perpendicular to a surface give rise to pressure on the surface

Example

A 4000kg elephant standing on one foot will cause a pressure of **250 000 Pa** under that foot.

What is the pressure on a pair of high heels shoes (area of 2.0 cm^2) for a lady of 55 kg?



FACT:

a woman in high heels is more deadly than an elephant

Alternative SI expressions of pressure

Force (N) = mass (kg) x acceleration (m/s²)

(Recall that weight = mass x gravitational acceleration)

Therefore Force also has a unit of kg m/s²

Pressure (Pa) = Force (N) /Area (m²)

$$\begin{array}{rcl}
 \text{Pa} & = & \frac{\text{kg m}}{\text{s}^2} \bigg| \frac{\text{m}^2}{\text{m}^2} \\
 & & \\
 & = & \frac{\text{kg}}{\text{s}^2} \bigg| \frac{\text{m}}{\text{m}}
 \end{array}$$

Pressure units

Pressure	$1 \text{ atm} = 1.01325 \times 10^5 \text{ N/m}^2 \text{ (Pa)} = 101.325 \text{ kPa}$ $= 1.01325 \text{ bar} = 760 \text{ mm Hg at } 0^\circ\text{C} = 10.333 \text{ m H}_2\text{O at } 4^\circ\text{C}$ $= 14.696 \text{ lb}_f/\text{in}^2 \text{ (psi)} = 33.9 \text{ ft H}_2\text{O at } 4^\circ\text{C}$
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- You can use the unit conversion method we learnt earlier to perform conversion for pressure units
- The blue font units are known as head of fluid. We will learn more about this way of representing pressure later on

Example

Convert 5.00 atm into psi

5.00 atm

14.696 psi

1 atm

= 73.5 psi

Factors for Unit Conversions

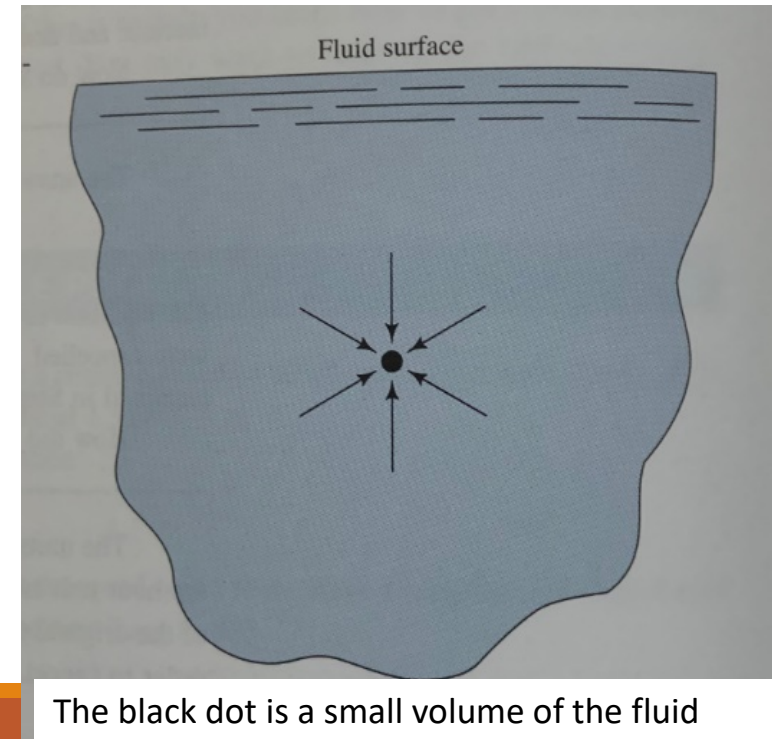
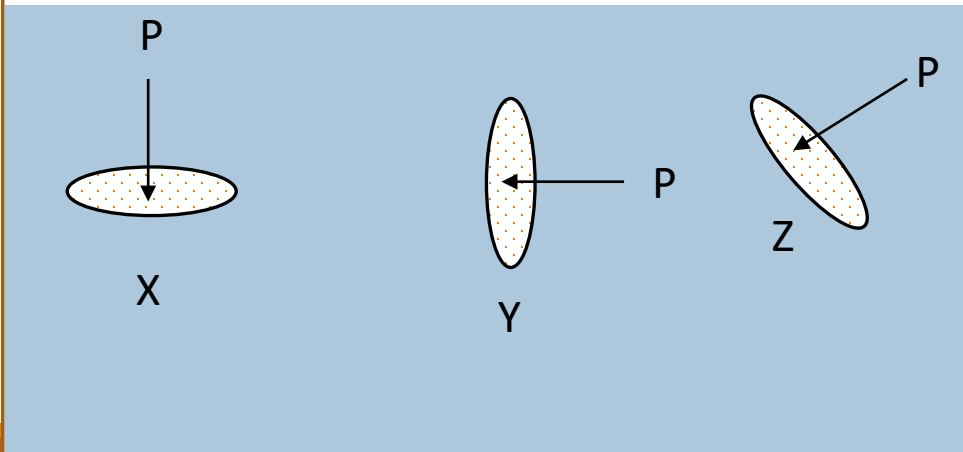
Quantity	Equivalent Values
Mass	1 kg = 1000 g = 0.001 metric ton = 2.20462 lb _m = 35.27392 ounces (oz) 1 lb _m = 16 oz = 453.593 g = 0.453593 kg
Length	1 m = 100 cm = 1000 mm = 10 ⁶ microns (μm) = 39.37 inches (in) = 3.2808 ft = 1.0936 yard (yd) = 0.0006214 mile 1 ft = 12 in = 1/3 yd = 0.3048 m = 30.48 cm
Volume	1 m ³ = 1000 L = 10 ⁶ cm ³ = 10 ⁶ ml = 35.3145 ft ³ = 264.17 gallon (gal) 1 ft ³ = 1728 in ³ = 7.4805 gal = 0.028317 m ³ = 28.317 litres (L) = 28,317 cm ³
Force	1 Newton (N) = 1 kg m/s ² = 0.22481 lb _f 1 lb _f = 32.174 lb _m ft/s ² = 4.4482 N
Pressure	1 atm = 1.01325 x 10 ⁵ N/m ² (Pa) = 101.325 kPa = 1.01325 bar = 760 mm Hg at 0°C = 10.333 m H ₂ O at 4°C = 14.696 lb _f /in ² (psi) = 33.9 ft H ₂ O at 4°C
Energy	1 Joule (J) = 1 N m = 2.778 x 10 ⁻⁷ kWh = 0.23901 calories (cal) = 9.486 x 10 ⁻⁴ British thermal unit (Btu)
Power	1 Watt (W) = 1 J/s = 0.23901 cal/s = 9.486 x 10 ⁻⁴ Btu/s = 1.341 x 10 ⁻³ hp
Temperature	T (°F) = 1.8 T(°C) + 32 T (K) = T(°C) + 273 T(°R) = T(°F) + 460 T(°R) = 1.8 T(K) F = Fahrenheit , C = Celsius , K = Kelvin , R = Rankine

Pressure – hydrostatic pressure

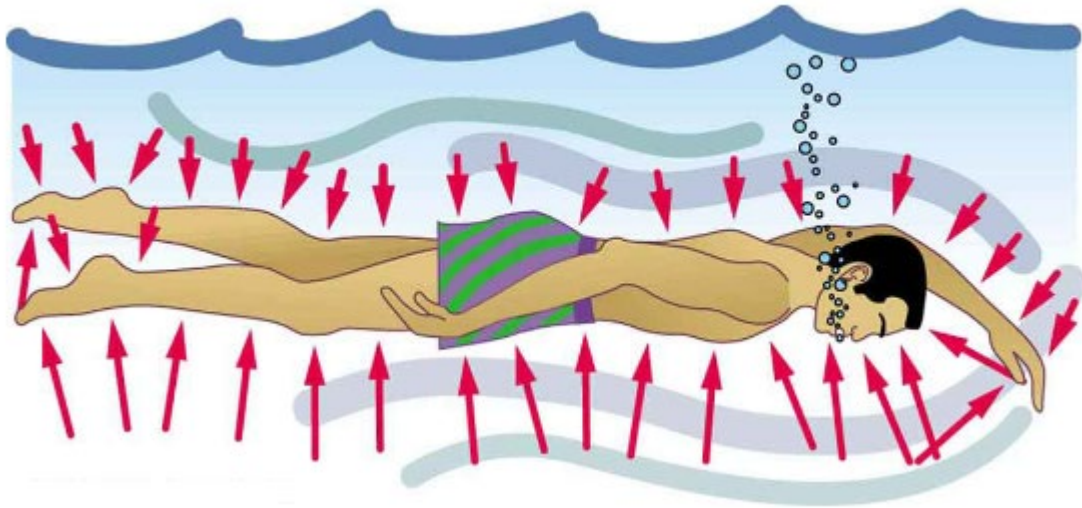
Pressure exerted by static fluid

- At any point in a fluid, pressure can act in any direction.
- Pressure acts uniformly in all directions on a small volume of fluid

The pressure P acts normally to the planes X, Y and Z, which represent small volume of fluid



Pressure exerted by static fluid

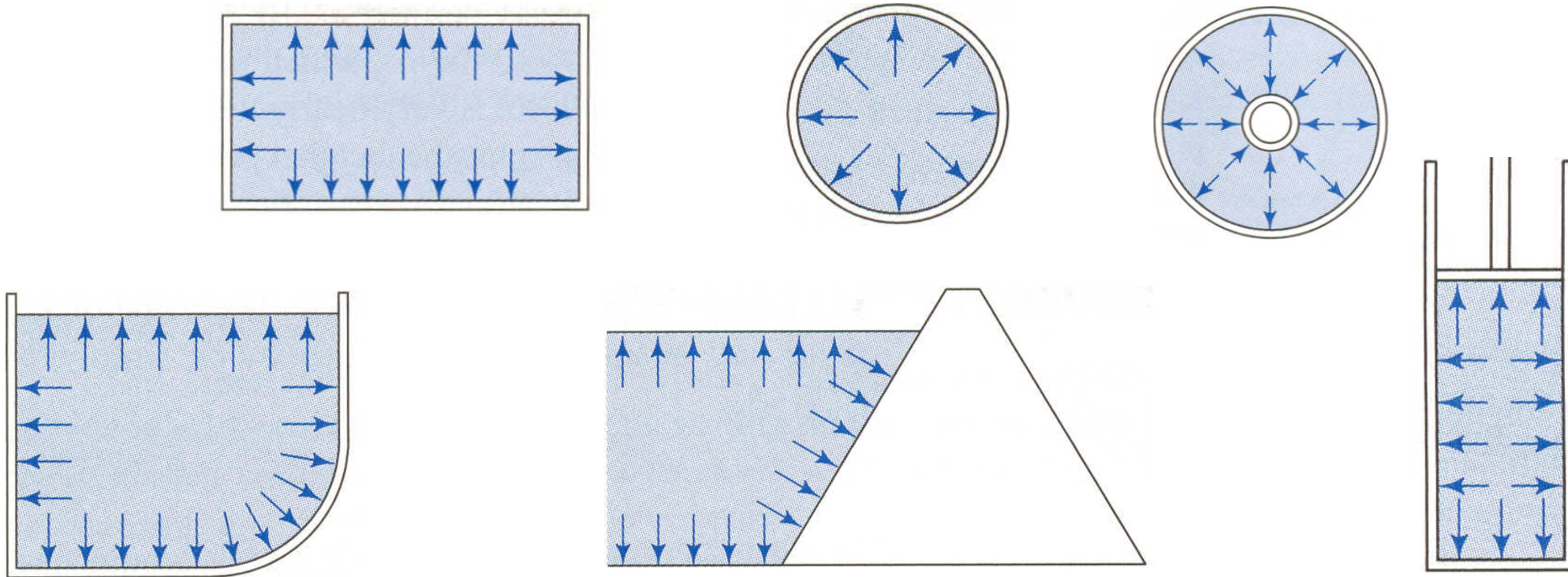


Source: Openstax Physics

- Pressure exerted by water on a swimmer
- Water would flow into the space he occupies if he were not there
- Arrow represents the direction and magnitude of forces exerted on swimmer
- Forces underneath swimmer are greater due to greater water depth – we will come to this shortly

Pressure exerted by static fluid

- In a fluid confined by solid boundaries, pressure acts perpendicular to the boundary

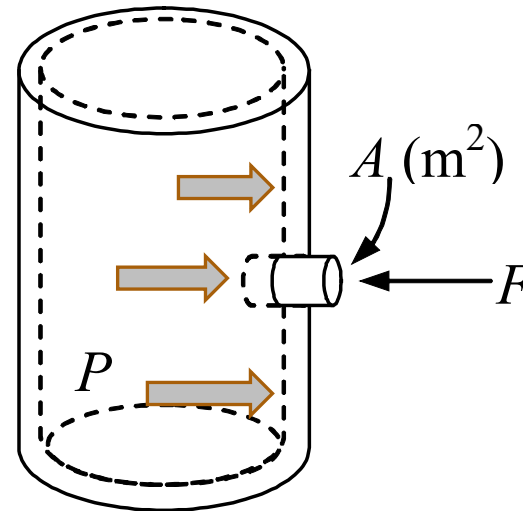


Pressure exerted by static fluid

Fluid pressure in a tank may be defined as the ratio F/A , where F is the minimum force that would have to be exerted on a frictionless plug in the hole to keep the fluid from emerging.

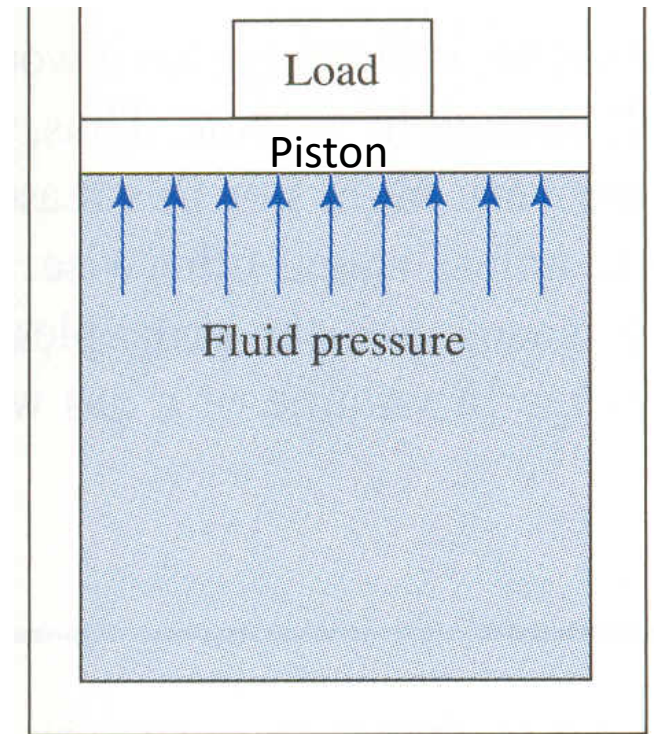
Fluid in the tank is exerting pressure on the wall of the tank as represented by the grey arrow.

Recall that In a fluid confined by solid boundaries, pressure acts perpendicular to the boundary



Example

Fluid pressure supports a load and a piston as shown in the diagram. Compute the magnitude of the fluid pressure. Given that the total weight of the piston and the load is $5.0 \times 10^2 \text{ N}$ and the area of the piston is 2500 mm^2 .



Hydrostatic pressure – pressure exerted by fluid



What pressure do you experience at the pool side?



As you dive down to the bottom of a swimming pool, how do you feel physically?

Hydrostatic pressure

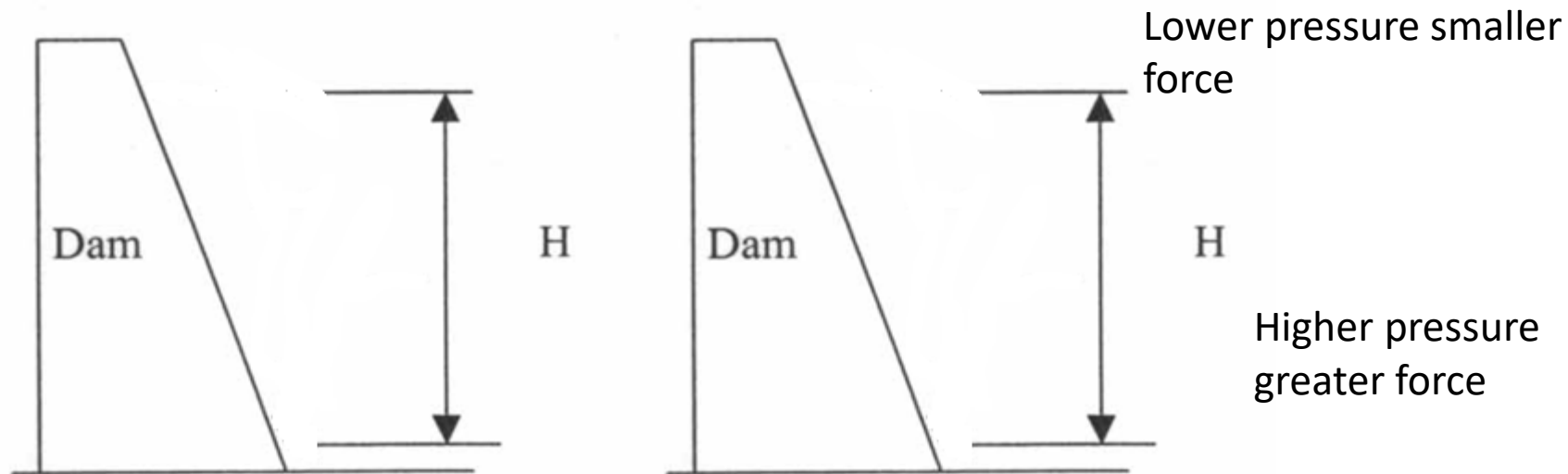
- Any object located below the fluid surface is subjected to pressure exerted by the fluid
- This pressure is
 - due to the weight of the fluid
 - called static fluid pressure
 - greater, the deeper you go

Atmospheric pressure is another example of pressure due to weight of a fluid (i.e. air in the atmosphere)



Hydrostatic pressure

- Thus design of a dam has to consider the greater force at the base of the dam.



Hydrostatic pressure

Equation derivation

Force at the surface = $P_o \times A$

Volume of fluid = $V = A \times h$

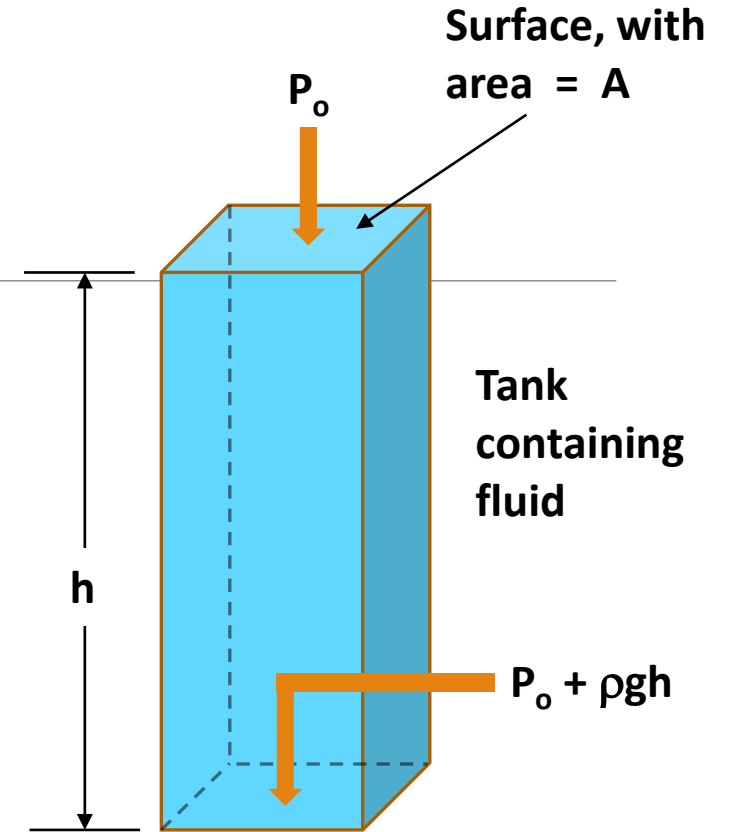
Mass of fluid = $m = \rho V$

Weight of fluid = $W = mg = (\rho V)g = \rho(Ah)g$

(this is the force exerted by weight of fluid on the bottom of tank)

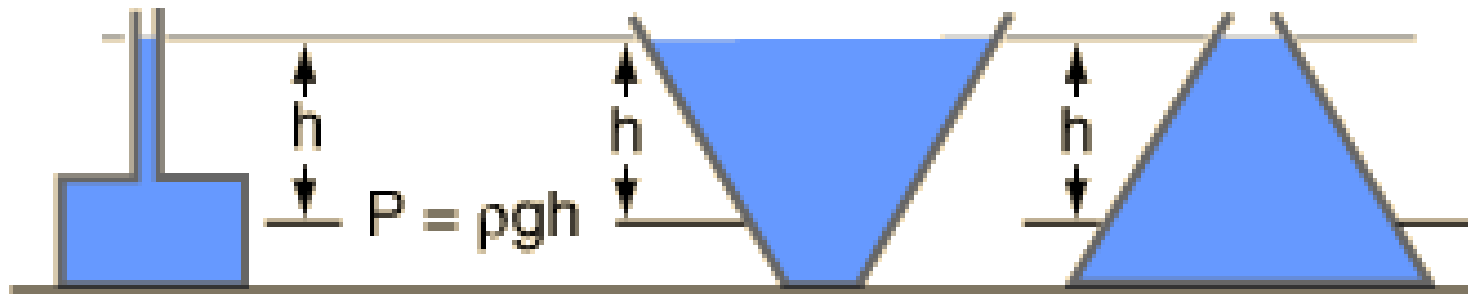
★ Total pressure at the bottom of the tank =

$$\frac{\text{Total Force}}{\text{Area}} = \frac{P_o A + \rho(Ah)g}{A} = \mathbf{P_o + \rho gh}$$



A = area
 h = height
 g = gravitational acceleration
 ρ = density of fluid
 P_o = pressure at the surface of fluid

Pressure exerted by liquid in a container does not depend on the shape of the container



The pressure exerted by the liquid at a depth of h is the same for all 3 containers above and can be calculated using the formula $P = \rho gh$

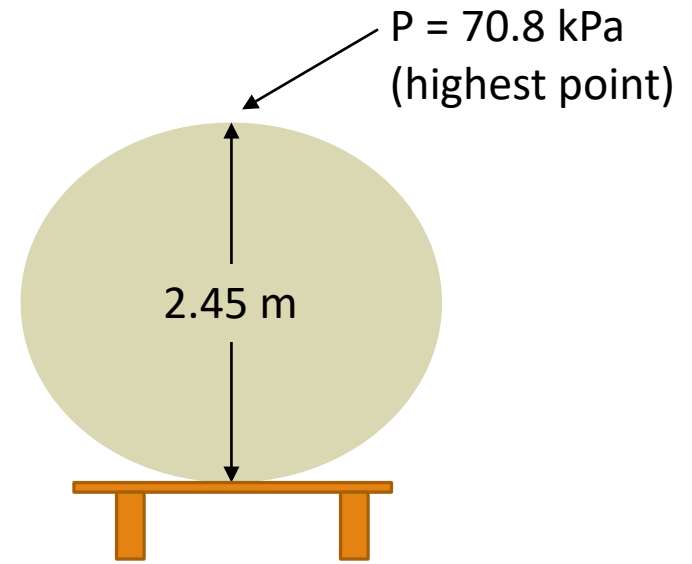
Example

Calculate the greatest pressure in a spherical tank containing olive oil. The tank is completely filled with olive oil and has a diameter of 2.45 m. The pressure measured at the highest point in the tank is 70.8 kPa. State your answer in 3 significant figures.

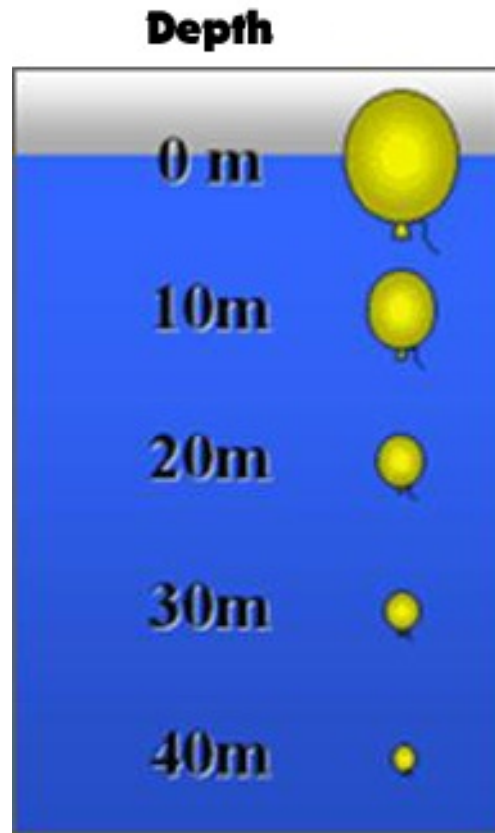
Given:

Density of olive oil = 920 kg/m^3

Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$



Example



What is the pressure difference between 0-10m?

Assumption:

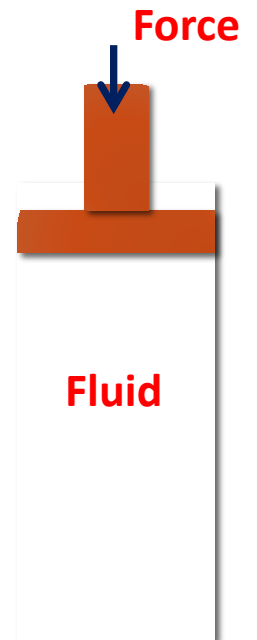
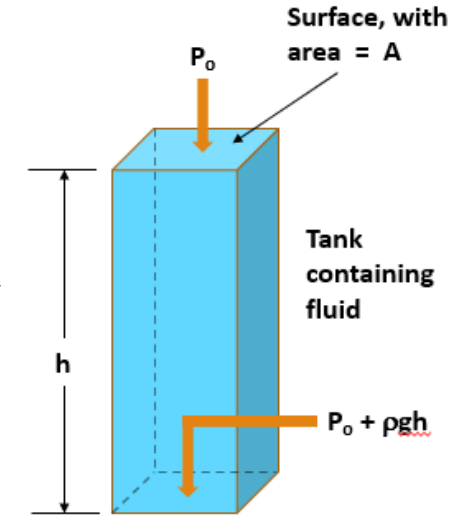
1. Density of fluid is 1000 kgm^{-3}
2. g is 9.81 m/s^2
3. The water surface is exposed to atmosphere and atmospheric pressure is 101.3 kPa

Pascal's Principle (Pascal's Law)

Hydrostatic pressure equation: $P = P_o + \rho gh$

Based on hydrostatic pressure equation:

- The pressure at the bottom of a container filled with liquid to a depth (h) is greater than pressure at surface (P_o) by " ρgh ". However, the pressure is the **SAME** at any 2 points at the **SAME LEVEL** in the fluid.
- If the pressure at surface (P_o) is increased using a piston that fits tightly inside the container to push down on the fluid surface, the pressure P at any depth increases by exactly the same amount. This is known as **Pascal's Principle** or **Pascal's Law**.
- Pascal's law: A change in pressure applied to an enclosed fluid is transmitted undiminished to all portions of the fluid and to the walls of its container.

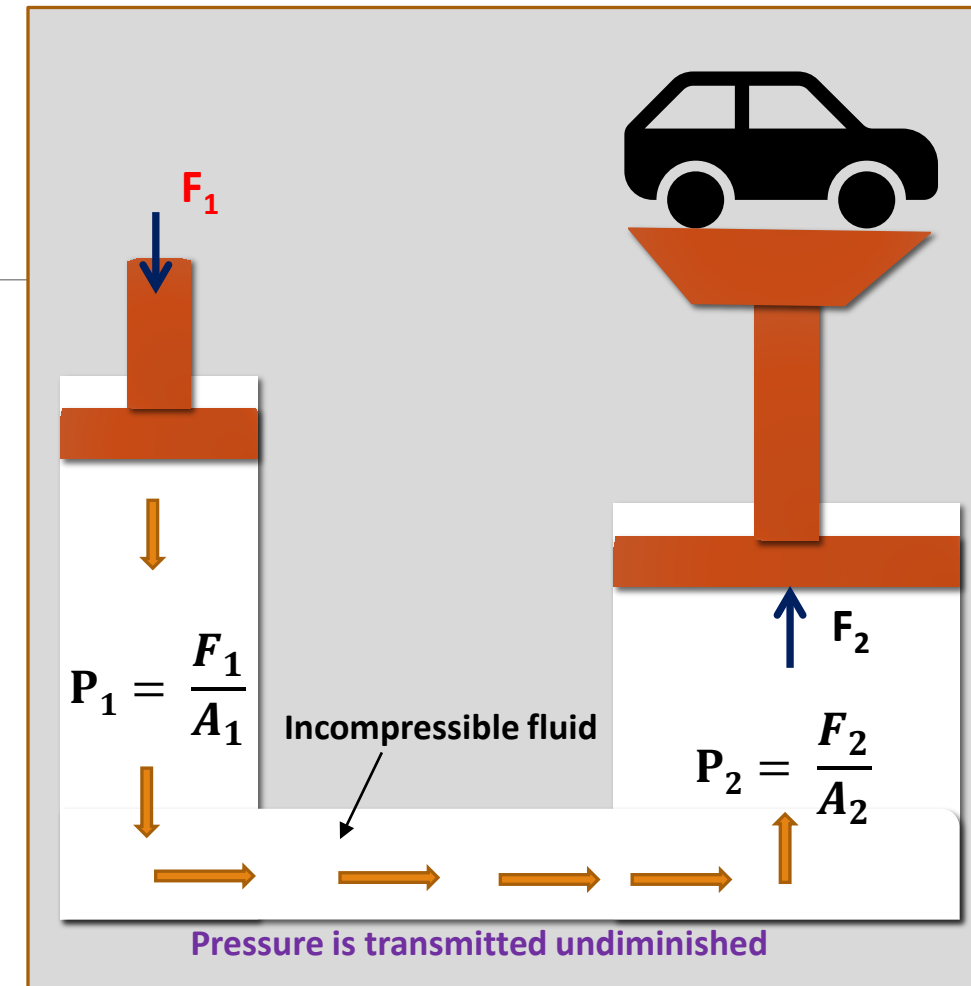


What happens when you squeeze a bottle?



Application of Pascal's Law

- Hydraulic system uses an incompressible fluid (e.g. oil) to transmit forces from one location to another within the fluid
- Recall Pascal's Law and $P = \text{Force}/\text{Area}$,
- $P_1 = P_2 \rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$
- Thus $F_2 = F_1 \times \frac{A_2}{A_1}$
- So, Pascal's law allows force to be multiplied by a factor of A_2/A_1 and a small force (F_1) exerted on the left piston is able to support the much bigger force (F_2)



P_1 = pressure created in the fluid due to force F_1
 P_2 = pressure experienced by the right piston. This is equal to P_1 due to Pascal's Law

Example

A hydraulic lift has a small cylindrical piston with radius of 5.0 cm and a larger piston with radius of 20 cm. The car with mass of 1,000 kg was place on the larger piston's platform.

- A. Assuming that the two pistons are at the same height, determine the force that must be applied to the small piston in order to lift the car?
- B. How far must the small piston move down to lift the car through a height of 0.10 m?

Assume $g = 9.81 \text{ m/s}^2$

In summary: equality of static fluid pressure at the same level for any U-shaped tube

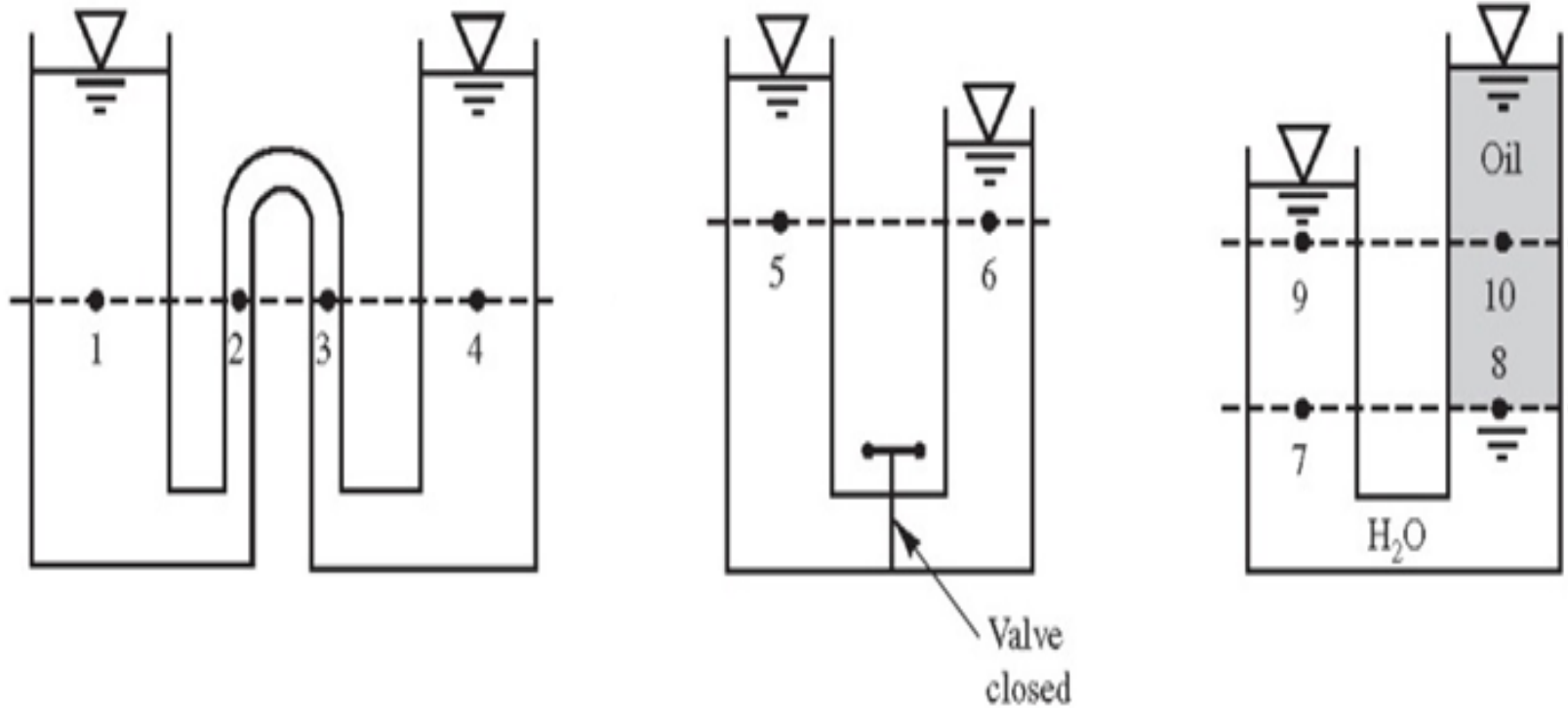
Pressure at C = Pressure at D
&
Pressure at E = Pressure at F

As long as:

1. same fluid
2. same elevation
3. fluid containing the points are connected

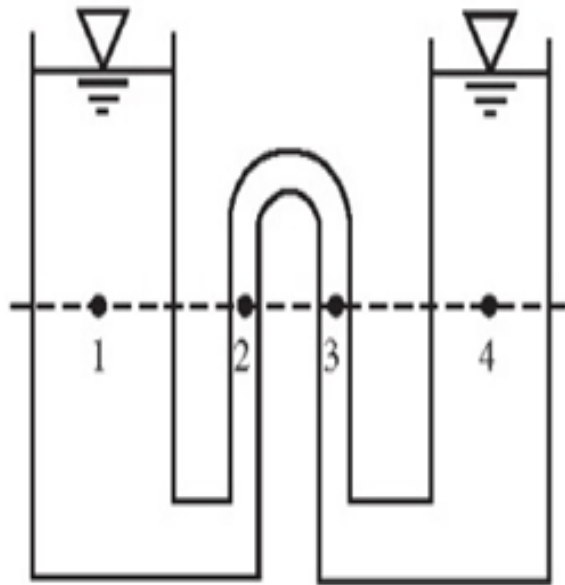


Example

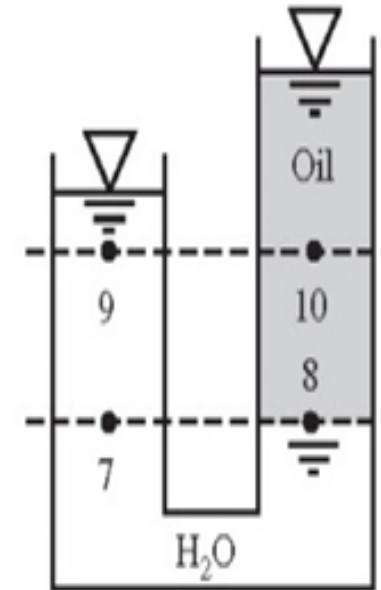
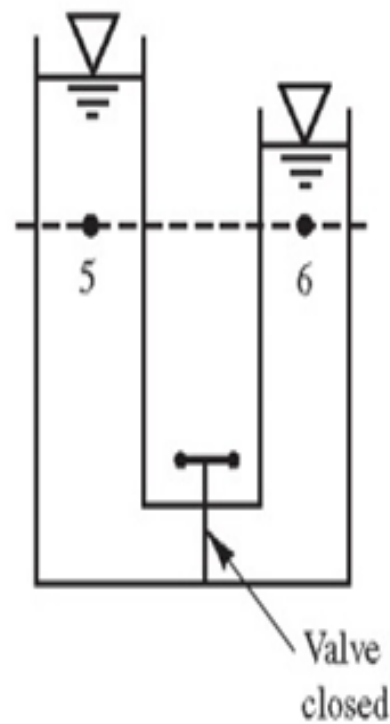


Which points have the same pressure?

Solutions



Point 1, 2, 3 and 4 have same pressure

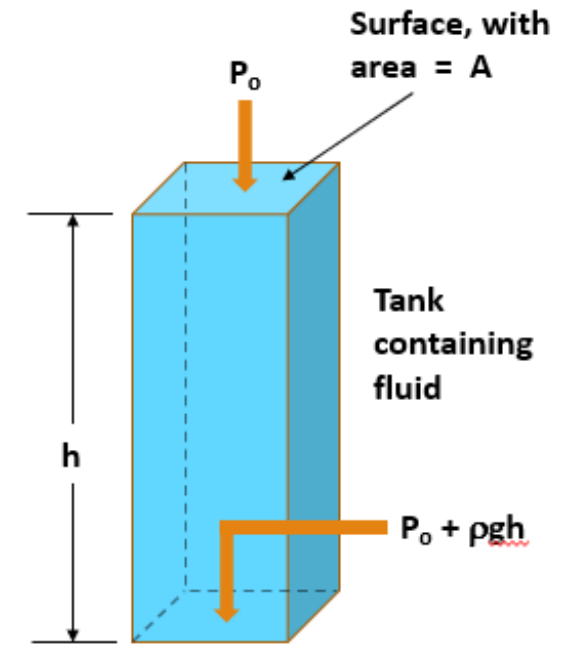


Point 7 and 8 have same pressure

Expressing pressure as head of liquid

Hydrostatic pressure equation: $P = P_o + \rho gh$

- From the hydrostatic pressure equation, pressure is sometimes expressed as the **head of liquid**
- Head of liquid: height of a hypothetical column of liquid if the pressure at the top was zero
- Example: 760 mm Hg → a pressure that is equivalent to that exerted by a column of mercury (Hg) that is of height 760 mm
- By specifying the height and type of fluid, we would be able to calculate the pressure using the equation ρgh



Example

What is 500 kPa in terms of mm Hg? Given that $g = 9.81 \text{ N/kg}$ and the density of mercury is 13600 kg/m^3 .